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① Stat 50

HW #8

Section 2.5

③ Mean of 1 = 900
σ of 1 = 30

1) μ of four :-

$$\mu = 900 \times 4 = 3600 \text{ hours}$$

$\mu = 3600$ hours

2) σ of four :-

$$4\sigma = 4\left(\frac{\sigma}{\sqrt{n}}\right) = 4\left(\frac{30}{\sqrt{4}}\right) = 4\left(\frac{30}{2}\right) = 15 \times 4$$

$\sigma = 60$

$\sigma = 60$ hours

⑧ Mean = 20.01 ounces
Std = 0.02 ounces.

(a) $\mu_x = 20.01$

$\Rightarrow \mu_{24} = 24 \times 20.01 = 480.24$

$\mu_{24} = 480.24$

(b) $\sigma_x = 0.02$

$$\sigma_{24\bar{x}} = 24 \cdot \left(\frac{0.02}{\sqrt{24}}\right) = 0.0979796$$

$$\boxed{\sigma_{24\bar{x}} = 0.09798}$$

(c) $$\boxed{\mu_{\bar{x}} = \mu = 20.01}$$

(d) $$\sigma_{\bar{x}} = \frac{\sigma_{24\bar{x}}}{24} = 24 \cdot \left(\frac{0.02}{\sqrt{24}}\right) \cdot \left(\frac{1}{24}\right)$$

$$\boxed{\sigma_{\bar{x}} = 0.004082483}$$

(e) [illegible]

$$\frac{0.02}{\sqrt{n}} = 0.0025$$

$$\Rightarrow \sqrt{n} = \frac{0.02}{0.0025} \Rightarrow n = \left(\frac{0.02}{0.0025}\right)^2$$

$$\boxed{n = 64}$$

There must be 64 bottles in a case for the standard deviation of the average volume per bottle to be 0.0025 ounces.

(2)

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(10)

Grade	revenue	Mean	Standard deviation
regular	2.60 /gal	1500	180
mid	2.75 /gal	500	90
Premium	2.90 /gal	300	40

(a) The mean daily revenue:-

$$DR = 2.60x_1 + 2.75x_2 + 2.90x_3$$

(a) $E[DR] = 2.60(1500) + 2.75(500) + 2.90(300)$

$$\boxed{E[DR] = 6145}$$

(b)

$$\sigma_{DR} = \sqrt{(2.60)^2(\sigma x_1)^2 + (2.75)^2(\sigma x_2)^2 + (2.90)^2(\sigma x_3)^2}$$

$$= \sqrt{6.75(180^2) + 7.5625(90^2) + 8.41(40^2)}$$

$$= \sqrt{218700 + 61256.25 + 13456}$$

$$= \sqrt{293412.25}$$

$$\boxed{\sigma_{DR} = 541.675}$$

(11) $\mu_x = 0.15 \frac{gal}{(mile\ pass)}$ $\quad \sigma_x = 0.01$

(a) $\mu_{210,8000x} = (210\ pass)(8000\ miles)(0.15)\frac{gal}{(pass)(miles)}$

(4)

$\mu = 252000$ gallons

(b) $\sigma_{\bar{x}} = \left(\frac{\sigma_x}{\sqrt{8000(210)}}\right) 8000(210)$

$\sigma_x = \frac{(0.01)(8000)(210)}{(\sqrt{8000})(\sqrt{2106})} = 12.9614814$

$\sigma_{\bar{x}} = 12.961$

(c) 0.15 gallons / 8000 mile and 210 passegers, so

fuel efficency :-

$\frac{0.15}{(8000)(210)} = \frac{x}{(8000)(210)}$

$x = 0.15$ gallons

(d) $\sigma_{\bar{x}} = 12.961$

then

$\frac{\sigma_{\bar{x}}}{8000(210)} = \frac{12.961}{(8000)(210)} = 0.0000007715$

$\sigma_{\frac{\bar{x}}{8000(210)}} = 7.7152 \times 10^{-6}$ gallons

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(17) Mean $= \mu_x = 40.25$

Std : $\sigma_x = 0.36$

(a) $\mu_x = 40.25$

$\mu_{\bar{x}} = 10(40.25)\left(\frac{1}{10}\right) = \frac{1}{n}(n)(\mu_x)$

$\boxed{\mu_{\bar{x}} = 40.25}$

(b) $\sigma_{\bar{x}} = \frac{\sigma_x}{\sqrt{n}} = \frac{0.36}{\sqrt{10}} = 0.113841996$

$\boxed{\sigma_{\bar{x}} = 0.11}$

(b) $\frac{0.36}{\sqrt{n}} = 0.05$

$\Rightarrow \sqrt{n} = \frac{0.36}{0.05}$

$\Rightarrow n = \left(\frac{0.36}{0.05}\right)^2$

$\Rightarrow n = 51.84$

Atleast 52 rocks must be sampled so that the standard deviation of sample mean SiO_2 content is 0.05.